

A Personal Formal-Discovery Conjecture and Its Settlement

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Abstract

This note records a single personal formal-discovery conjecture and a formal result that settles it within the stated AMLD/Board framework. The result gives an exact necessary-and-sufficient condition for when certified pre-outcome self-state can support a verifier-safe adaptive policy that strictly improves expected verified settlement cost over the best fixed verifier-safe policy. No claim of novelty relative to the mathematical literature is made here.

1 Personal Formal-Discovery Conjecture

Conjecture 1 (Personal Formal-Discovery Conjecture). *I will make a formalized discovery pertaining to the necessary and sufficient conditions under which a Board-controlled AMLD can use certified pre-outcome self-state to identify a decision region where switching from its default verifier-safe proof policy to an alternate verifier-safe policy strictly lowers expected verified settlement cost, without altering the verifier trust boundary and without sacrificing eventual complete fallback when that fallback is retained by the operational design.*

The following theorem supplies such a characterization.

2 Decision-Region Characterization

Let Π be a finite family of verifier-safe search policies. Let Σ be certified information available before the policy for the current episode is selected, and let

$$\mathcal{G} = \sigma(\Sigma)$$

be the information sigma-algebra generated by that certified self-state. For each $\pi \in \Pi$, let L_π denote the total integrable verified settlement cost incurred by using π , where all declared switching, control, translation, or other policy-selection overhead is charged on the same cost accounting basis.

For each $\pi \in \Pi$, choose a $\mathcal{G}$ -measurable version

$$G_\pi = \mathbb{E}[L_\pi \mid \mathcal{G}]$$

of its conditional expected cost. The best fixed-policy cost is

$$J_{\text{fix}} = \min_{\pi \in \Pi} \mathbb{E}[L_\pi].$$

Let M range over all $\mathcal{G}$ -measurable selectors with values in Π . For such a selector define

$$L_M = \sum_{\pi \in \Pi} \mathbf{1}_{\{M=\pi\}} L_\pi,$$

and define the best adaptive Board cost by

$$J_{\text{adapt}} = \inf_M \mathbb{E}[L_M].$$

Theorem 1 (Decision-Region Characterization). *Assume Π is finite and every L_π is integrable. Then*

$$J_{\text{adapt}} = \mathbb{E} \left[\min_{\pi \in \Pi} G_\pi \right] \leq J_{\text{fix}}.$$

Choose any best fixed policy

$$\pi^* \in \arg \min_{\pi \in \Pi} \mathbb{E}[L_\pi]$$

and define the conditional decision advantage

$$\Delta = G_{\pi^*} - \min_{\pi \in \Pi} G_\pi.$$

Then $\Delta \geq 0$ almost surely, and the following are equivalent:

1. *some $\mathcal{G}$ -measurable Board selector strictly beats the best fixed-policy cost;*
2. *$J_{\text{adapt}} < J_{\text{fix}}$;*
3. *$\mathbb{E}[\Delta] > 0$;*
4. *$\mathbb{P}(\Delta > 0) > 0$.*

Thus a strict adaptive improvement exists if and only if certified pre-outcome self-state reveals a positive-probability decision region

$$D = \left\{ G_{\pi^*} > \min_{\pi \in \Pi} G_\pi \right\}$$

on which some verifier-safe policy has strictly lower conditional total cost than a best fixed verifier-safe policy.

Proof. Because Π is finite and each G_π is $\mathcal{G}$ -measurable, the pointwise minimum

$$G_{\min} = \min_{\pi \in \Pi} G_\pi$$

is $\mathcal{G}$ -measurable. Fix an ordering of Π and, at each outcome, let M^* be the first policy in that ordering that attains $G_{\min}$. Finiteness of Π implies that M^* is $\mathcal{G}$ -measurable.

For any $\mathcal{G}$ -measurable selector M ,

$$\begin{aligned} \mathbb{E}[L_M | \mathcal{G}] &= \sum_{\pi \in \Pi} \mathbf{1}_{\{M=\pi\}} \mathbb{E}[L_\pi | \mathcal{G}] \\ &= \sum_{\pi \in \Pi} \mathbf{1}_{\{M=\pi\}} G_\pi \\ &\geq G_{\min} \end{aligned}$$

almost surely. For $M = M^*$ equality holds almost surely. Taking expectations gives

$$J_{\text{adapt}} = \mathbb{E}[L_{M^*}] = \mathbb{E}[G_{\min}] = \mathbb{E} \left[\min_{\pi \in \Pi} G_\pi \right].$$

Thus the adaptive infimum is actually attained.

For every fixed policy π ,

$$\mathbb{E}[L_\pi] = \mathbb{E}[G_\pi] \geq \mathbb{E}[G_{\min}],$$

by the tower property. Hence $J_{\text{adapt}} \leq J_{\text{fix}}$.

Now fix

$$\pi^* \in \arg \min_{\pi \in \Pi} \mathbb{E}[L_\pi].$$

By definition,

$$\Delta = G_{\pi^*} - G_{\min} \geq 0$$

almost surely. Again using the tower property,

$$\begin{aligned} J_{\text{fix}} - J_{\text{adapt}} &= \mathbb{E}[L_{\pi^*}] - \mathbb{E}[G_{\min}] \\ &= \mathbb{E}[G_{\pi^*}] - \mathbb{E}[G_{\min}] \\ &= \mathbb{E}[\Delta]. \end{aligned}$$

Therefore

$$J_{\text{adapt}} < J_{\text{fix}} \iff \mathbb{E}[\Delta] > 0.$$

Since $\Delta \geq 0$ almost surely and is integrable,

$$\mathbb{E}[\Delta] > 0 \iff \mathbb{P}(\Delta > 0) > 0.$$

Finally, M^* attains J_{adapt} , so $J_{\text{adapt}} < J_{\text{fix}}$ holds if and only if some $\mathcal{G}$ -measurable selector has expected cost strictly below the best fixed-policy cost. This proves all four equivalences. $\square$

Remark 1 (Pointwise self-state notation). *If the self-state takes values in a standard Borel space, one may choose regular conditional expectations and write*

$$g_\pi(s) = \mathbb{E}[L_\pi \mid \Sigma = s]$$

for suitable measurable versions. Then $G_\pi = g_\pi(\Sigma)$ almost surely, and the theorem reduces to the intuitive pointwise formulation

$$J_{\text{adapt}} = \mathbb{E} \left[\min_{\pi \in \Pi} g_\pi(\Sigma) \right].$$

The sigma-algebra formulation above is used in the theorem so that no point-conditioning assumption is needed.

3 Settlement of the Conjecture

The theorem supplies the requested necessary-and-sufficient condition. The formal boundary between justified adaptive switching and unjustified switching is not merely that the self-state is predictive. Rather, the self-state must induce a positive-probability region on which the conditional total cost of an alternate verifier-safe policy is strictly below that of a best fixed verifier-safe policy, after every declared overhead has been charged.

Equivalently,

$\text{strict adaptive improvement} \iff \mathbb{P}(\Delta > 0) > 0$

under the hypotheses of the theorem.

Soundness is orthogonal to the cost comparison: if every policy in Π remains verifier-gated, adaptive selection does not enlarge the set of accepted certificates. Eventual complete fallback is likewise an operational condition: it is preserved when the Board’s scheduling design guarantees a fair complete fallback channel or restricts alternate policies to bounded excursions that return to such a channel.

Therefore, within the formal framework stated above, the Decision-Region Characterization Theorem settles the adaptive-cost component of the Personal Formal-Discovery Conjecture. The verifier-safety and eventual-complete-fallback components follow under the separately stated verifier-gating and fallback-design assumptions. Under those assumptions together, the Personal Formal-Discovery Conjecture is settled positively.